

NIKHIL DEEKONDA

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EDUCATION

Yeshiva University, Katz School of Health and Science, NY, USA

Sep 2023 - May 2025

Masters of Science in Artificial Intelligence

CGPA 3.9

WORK EXPERIENCE

Valuai.io, Los Angeles, California, USA

May 2025 – Present

AI Engineer

- Architected an AI-driven patient engagement platform integrating AWS Bedrock, LangChain, Redis, and MongoDB for secure, scalable clinician–patient interactions and intelligent health data management.
- Designed a Mixture of Experts (MoE) architecture on top of a DeepSeekR1-distilled LLaMA 8B model, introducing domain-specific experts for different plans involved, each trained via RAG-based domain datasets.
- Engineered a RAG pipeline with HNSW indexing for dense retrieval, cross-encoder reranking for contextual relevance, and a unified retrieval layer serving all domain experts for factual, and clinically grounded inference.
- Currently building a Neo4j knowledge graph to enhance structured reasoning and relationship-based querying, improving clinical recommendation accuracy and enabling better longitudinal patient insights.
- Fine-tuned large language models on the patient-side interface using synthetic conversational data generated via few-shot prompting with GPT-4o and DeepSeek, aligning outputs with clinical guidelines and ensuring safe, context-aware multi-turn interactions.
- Developed and deployed backend microservices using FastAPI, containerized with Docker and orchestrated on AWS EC2, integrating AWS Cognito for secure authentication and Redis for low-latency session retrieval.
- Deployed fine-tuned models on RunPod and FAL for cost-efficient, low-latency inference at production scale.

LTIMindtree, Pune, Maharashtra, India

Jul 2022 - Jul 2023

Senior Software Engineer

- Developed guardrails, data pipelines, and validation flows for ML/DL models including LLMs; fine-tuned models like GPT, Falcon, LLaMA-2, and Mistral for real-world use cases (PII removal, tone classification, toxic word detection) using LoRA and QLoRA techniques for production-scale deployments.
- Built a scalable platform to fine-tune LLMs with multiple techniques and implemented output validation for LLM agents utilizing frameworks such as AutoGen and LangGraph to ensure reliable and consistent responses.
- Designed and executed complex Retrieval-Augmented Generation (RAG) workflows integrating LLMs with vector databases; developed hallucination detection systems leveraging Small Language Models (SLMs).
- Trained a proprietary 350M parameter language model from scratch and deployed fine-tuning, RAG, and validation workflows on AWS using services like EC2, S3, SageMaker, and Bedrock for enterprise-grade scalability.

National Highway Authority of India, Tirupati, Andhra Pradesh, India

Jun 2021 - Aug 2021

Machine Learning Intern

- Applied various machine learning algorithms to NHAI road dataset of half a million instances, evaluating models based on R-squared values, successfully predicting road rutting factors, based on the best performing model.
- Conducted model evaluation and feature analysis to identify key predictors, enabling data-driven insights for road quality and maintenance planning.
- Engineered data preprocessing and feature engineering pipelines using Python and Scikit-learn, optimizing model performance and reducing prediction error for rutting analysis by over 15%.

PROJECTS

Mixture-of-Experts (MoE) Implementation in PyTorch

- Developed and optimized advanced Mixture-of-Experts (MoE) architectures in PyTorch, supporting both single-device and multi-device distributed environments for scalable LLM research and experimentation.
- Engineered three MoE variants-Basic, Sparse, and Shared Expert Sparse MoE—demonstrating expertise in dynamic routing, expert specialization, and model scalability principles.
- Implemented efficient token-level dynamic gating using top-*k* selection and tensor-based dispatch mechanisms to route inputs to the most relevant experts, optimizing compute utilization.
- Designed and integrated a hybrid load-balancing loss combining auxiliary loss and z-loss, achieving uniform expert utilization and reducing training instability by 25%.
- Deployed multi-device distributed training using PyTorch DistributedDataParallel, achieving a 40% improvement in training efficiency and maintaining expert synchronization across GPUs and nodes.

TaskFin: RAG-Based Financial Task Automation

- Designed and implemented a RAG-based system for end-to-end bill payment automation, leveraging LangChain agents and the LangGraph framework to orchestrate multi-turn interactions and ensure context-aware, accurate financial task management from natural language input.
- Developed a robust Orchestrator Agent to coordinate specialized sub-agents for Authentication, Financial Transactions, and State Management, using LangChain memory (buffer, entity, summary) and MCP servers to maintain persistent contextual state across interactions.
- Designed detailed PRDs (Product Requirement Documents) before initiating development, ensuring clear objectives and measurable outcomes for the project.
- Deployed the complete solution as a fully interactive Streamlit conversational app, enabling real-world RAG-driven task automation, secure data handling, and rapid prototyping of AI-powered fintech assistants.

LungAware: AI Lung Cancer Detection App

- Engineered a deep convolutional neural network (CNN) for early lung cancer detection, achieving 99.4% classification accuracy through advanced image preprocessing, transfer learning, and hyperparameter optimization.
- Developed cross-platform (Android/iOS) mobile applications with Swift (iOS) and Kotlin (Android), integrating TFLite-quantized models and Grad-CAM for explainable AI and transparent diagnostic support.
- Implemented a seamless user interface and cloud-based backend (AWS EC2) for automated inference, patient management, and result notification, ensuring compliance with healthcare privacy and security best practices.

SKILLS

Programming Languages	Python, Swift, Java, JavaScript, R, C, C++
Frameworks / Libraries	PyTorch, TensorFlow, FastAPI, LangChain, CrewAI, LangGraph, CoreML
Databases	PostgreSQL, MySQL, MongoDB, Redis
Cloud / DevOps	AWS, GCP, Docker, Git, GitHub
Tools / Analytics	Excel, Tableau, Jupyter, VS Code, XCode
Other	LLM Fine-tuning, RAG Pipelines, Multi-agent Systems, iOS Development (SwiftUI)

CERTIFICATIONS

- [AWS Certified Machine Learning Engineer – Associate](#)
- [AWS Certified AI Practitioner](#)
- [Engineer Data in Google Cloud](#)
- [Exploring Data With Looker](#)